

Exercise 5: Triggers

Scenario 1: Automatically update the last modified date when a customer's record is updated.

Question: Write a trigger **UpdateCustomerLastModified** that updates the LastModified column of the Customers table to the current date whenever a customer's record is updated.

Program

```
CREATE OR REPLACE TRIGGER UpdateCustomerLastModified
```

```
BEFORE UPDATE
```

```
ON Customers
```

```
FOR EACH ROW
```

```
BEGIN
```

```
    :NEW.LastModified := SYSDATE;
```

```
END;
```

Scenario 2: Maintain an audit log for all transactions.

Question: Write a trigger **LogTransaction** that inserts a record into an AuditLog table whenever a transaction is inserted into the Transactions table.

Program

```
CREATE TABLE AuditLog
```

```
(
```

```
    AuditID NUMBER GENERATED BY DEFAULT AS IDENTITY,
```

```
    TransactionID NUMBER,
```

```
    ActionDate DATE,
```

```
    Action VARCHAR2(50)
```

```
);
```

CREATE OR REPLACE TRIGGER LogTransaction

AFTER INSERT

ON Transactions

FOR EACH ROW

BEGIN

 INSERT INTO AuditLog

 (

 TransactionID,

 ActionDate,

 Action

)

VALUES

 (

 :NEW.TransactionID,

 SYSDATE,

 'Transaction Inserted'

);

END;

Scenario 3: Validate Deposits and Withdrawals

Scenario 1: Automatically update the last modified date when a customer's record is updated.

Question: Write a trigger **UpdateCustomerLastModified** that updates the LastModified column of the Customers table to the current date whenever a customer's record is updated.

Program

CREATE OR REPLACE TRIGGER CheckTransactionRules

BEFORE INSERT

ON Transactions

FOR EACH ROW

DECLARE

v_balance NUMBER;

BEGIN

SELECT Balance

INTO v_balance

FROM Accounts

WHERE AccountID = :NEW.AccountID;

IF :NEW.TransactionType = 'Withdrawal'

AND :NEW.Amount > v_balance THEN

RAISE_APPLICATION_ERROR(

-20001,

'Insufficient balance for withdrawal.'

);

END IF;

IF :NEW.TransactionType = 'Deposit'

AND :NEW.Amount <= 0 THEN

RAISE_APPLICATION_ERROR(

-20002,

'Deposit amount must be greater than zero.'

);

END IF;

END;